

HomeWork #1 Due Sep 9

1.2	Page 2	1a,b,e,f, 2a,d,e,f, 3, 4c, 5-6
1.3	Page 4	1, 3
1.4	Page 5	1-3, 5, 7
1.6	Page 10	2, 4, 5
2.1	Page 13	2, 7, 10e, 11
2.2	Page 17	3, 4

- P. 1. Prove there does not exist an order relation $<$ on $\mathbb{C}$ such that $(\mathbb{C}, +, \cdot, <)$ is an ordered field.

HomeWork #2 Due Sep 23

2.3	Page 20	1, 3-6, 8
2.5	Page 28	1, 3-6, 10
3.1	Page 33	2-7

- P. 1. Find the loci of points satisfying:

a) $\operatorname{Re} \frac{1}{z} > \frac{1}{2}$

b) $|z^2 - 1| = \alpha, \alpha > 0$

- P. 2. Let $M = \{x \mid 0 \leq x \leq 1, x = 0.x_1x_2x_3x_4x_5 \cdots, x_i \text{ odd}\}$ i.e., M is the set of numbers between 0 and 1 (inclusively) with infinite decimal representations all of whose digits are odd.

Question. Is M closed?

- P. 3. Suppose $\{z_n\} \rightarrow \zeta$. Show $z'_n = \frac{z_1 + z_2 + \cdots + z_n}{n} \rightarrow \zeta$.

Homework #3 Due Oct 14

3.1	Page 33	2-7
3.2	Page 43	1, 3-4, 6-12, 14-15, 17, 19, 21

Homework #4 Due Oct 30

3.3 Page 54 1, 3, 5, 8, 10, 13-18, 20, 22-23, 28, 30

P. 1. Let $f(z) = \exp(-1/|z|)$. Show that this function is uniformly continuous on $D = \{z : 0 < |z| < 1\}$.

P. 2. Show that $e^z > 1 + z$ for $z \in \mathbb{R}, z \neq 0$.

P. 3. Find all solutions of:

a) $\cos 2z = 3i$

b) $\sin z = 8$

P. 4. The domain $\{z : |z| < 1\}$ is mapped onto the upper half-plane by a bi-linear transformation which takes $1, i, -1$ into $0, 1, \infty$, respectively. Find the mapping. What are the images of the radii of the unit circle leading (from 0) to the points $1, i, -1, -i$?

P. 5. Let $D = B(0,1)$ and $E = B(0,1) \setminus \overline{B}(-1/2, 1/2)$. Find the unique one-to-one, conformal mapping $f : D \rightarrow E$, $f(0) = 1/2$, $f'(0) > 0$.

P. 6. Let $D = B(0,1)$ and $E = B(0,1) \setminus (-1, -1/2]$. Find the unique one-to-one, conformal mapping $f : D \rightarrow E$, $f(0) = 0$, $f'(0) > 0$.

Homework #5 Due Nov 18

4.1 Page 67 1, 5-7, 8-13, 16, 19-22

4.2 Page 73 1-5, 7a,c,d, 8, 9a-d, 10-11, 13

Homework #6 Due Dec 2

4.3 Page 80 1-2, 4, 6-10

4.5 Page 87 1, 3, 4, 6-9

Homework #7 Not Assigned

4.6	Page 95	4-6, 8, 10-11
4.7	Page 99	2-4, 6-7
5.1	Page 110	1a,b,c,e,h,j, 4, 6, 8, 10, 13-14, 16
5.2	Page 121	1,a,c, 2,a,b,c,d, 3-4, 6
5.3	Page 126	2, 6, 9-10

P. 1. Verify the parenthetical comment on page 98:

To show the second equality above takes a little effort, although for γ smooth it is easy. The details are left to the reader.